

Tuesday, April 14

- Midterm next Thursday (4/23)

- No class next Tuesday (4/21)

- Office hours:

 - ↳ Canceled next Thursday (4/23)

 - ↳ Virtual Monday (4/20)

 - ↳ Extra tomorrow (4/15)

↙ If second midterm grade is higher,
I will replace first midterm grade.

Plan

Reinforcement learning!

Up to now, we assumed
data is static and
learning is off line but...

- ↳ playing games
- ↳ autonomous movement
- ↳ self-driving
- ↳ trading
- ↳ chatting

Challenge: Predictions
influence
future data

⇒ Reinforcement learning
(AlphaGo, DOTA 2 AI, ChatGPT)

Motivation: Natural intelligence

- ↳ infant
- ↳ 2-year old
- ↳ middle school
- ↳ college

In state, take action, get reward, update state

Temple Run

State

Pixels

Action

Left/right/jump/slide

Reward

Alive + coins

Update

Next pixels



Stock Market

Chat

Mathematical Formulation

Time steps $t=1, 2, \dots, T$

At time t ,

- State s_t
 - Take action a_t
 - Get reward $r(s_t, a_t)$
 - Update $s_{t+1} = f(s_t, a_t)$
- } Can all be stochastic!

Trajectory $\tau = s_1, a_1, s_2, a_2, \dots, s_T, a_T$

$$R(\tau) = - \sum_{t=1}^T r(s_t, a_t) \gamma^{t-1}$$

γ ← discount future reward
 $0 < \gamma < 1$

Model

Policy π_θ : state $\rightarrow$ action dist.

$$a_t \overset{*}{\sim} \pi_\theta(s_t)$$

Loss

$$\begin{aligned} \mathcal{L}(\theta) &= \mathbb{E}_{\tau \sim \pi_\theta} R(\tau) \\ &= \sum_{\tau} \pi_\theta(\tau) R(\tau) \end{aligned}$$

$\uparrow$
prob. of τ

In practice, sample τ and approximate $\mathcal{L}$.

* If deterministic, could memorize e.g., Atari-style games

Policy Gradient

$$\mathcal{L}(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} R(\tau) = \sum_{\tau} \pi_{\theta}(\tau) R(\tau) \approx \frac{1}{m} \sum_{\ell=1}^m \pi_{\theta}(\tau_{\ell}) R(\tau_{\ell})$$

estimate

$$\nabla_{\theta} \mathcal{L}(\theta) = \sum_{\tau} R(\tau) \nabla_{\theta} \pi_{\theta}(\tau)$$

can compute where's the expectation? using back prop but

Goal: We want $\nabla_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} R(\tau)$ to be expectation so we can estimate

Hint: Compute $\nabla_{\theta} \log \pi(\tau) = \dots$

Then $\nabla_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} R(\tau) = \dots$

REINFORCE Algorithm

For $l=1, \dots, m$:

Sample $\tau = (s_1, a_1, \dots, s_T, a_T)$ from π_θ

Compute $R(\tau) = \sum_{t=1}^T -r(s_t, a_t) \gamma^{t-1}$

$\theta \leftarrow \theta - \alpha R(\tau) \nabla_{\theta} \log \pi(\tau)$

Note that:

↳ reward can be non-differentiable

↳ we can extract more trajectories

↳ Right in expectation but variance can be high ... subtract baseline
(Also view as "centering" neural reward)

$\left\{ \begin{array}{l} \tau_1 = (s_1, a_1, \dots, s_T, a_T) \\ \tau_2 = (s_2, a_2, \dots, s_T, a_T) \\ \vdots \end{array} \right.$